Heat regulation

Honeybees are renowned for controlling the temperature of their hive. When it gets too hot, they ventilate it by using their wings to fan the hot air out of the nest. When it gets too cold, the bees generate metabolic heat by rapidly contracting and relaxing their flight muscles. They also use heat as a defense mechanism. Japanese giant hornets are fierce predators of honeybees. Capable of killing large numbers quickly, they pose a serious threat to bees' nests. Hornets begin their attacks by picking off single honeybees at the entrance to the hive. However, Japanese honeybees defend themselves with self-generated heat. If a hornet attacks, they swarm around it, vibrating their wings to raise their collective temperature. Since the hornet cannot tolerate a temperature above 114.8°F (46°C) whereas the bees can survive at almost 118.4°F (48°C), the attacker eventually dies.